KING SALMAN INTERNATIONAL UNIVERSITY

Field of ENGINEERING

Engineering Program
All Engineering Program

MAT121 Statics

3 Cr. Hrs. = (***2*** LCT + ***2*** TUT + ***0*** LAB + ***0*** OTH)

Lecture : (Vectors for mechanics)
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Date : 13 / 10 / 2026

Cartesian Coordinate System

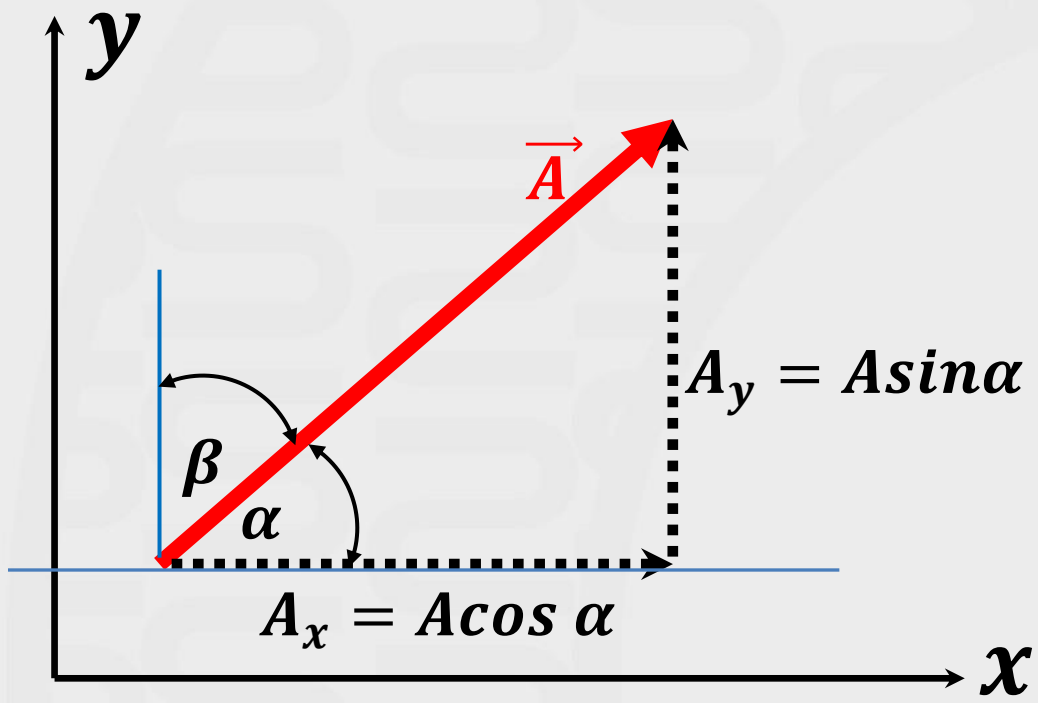
Vector Components Two-dimensions

Any vector in two dimensions can be broken down into two components along the x and y axes. Assume a vector $\vec{A}$ as shown in Figure is broken up or resolved into two components vectors.

$$\vec{A} = A\cos\alpha\, \boldsymbol{i} + A\sin\alpha\, \boldsymbol{j}$$

OR

$$\vec{A} = A\cos\alpha\, \boldsymbol{i} + A\cos\beta\, \boldsymbol{j}$$



4 STEPS For Solution



Cartesian Form

Convert each force to Cartesian form

$$\overrightarrow{\boldsymbol{F}} = F\cos\alpha\,\boldsymbol{i} + F\sin\alpha\,\boldsymbol{j}$$



Resultant Vector

i Components
j components



Resultant Magnitude

The magnitude of the vector

$$R = \sqrt{R_x^2 + R_y^2}$$



Resultant Dir.

$$\theta = \tan^{-1}\left|\frac{R_y}{R_x}\right|$$

Vector Components Three-dimensions

Cartesian form of the vector:

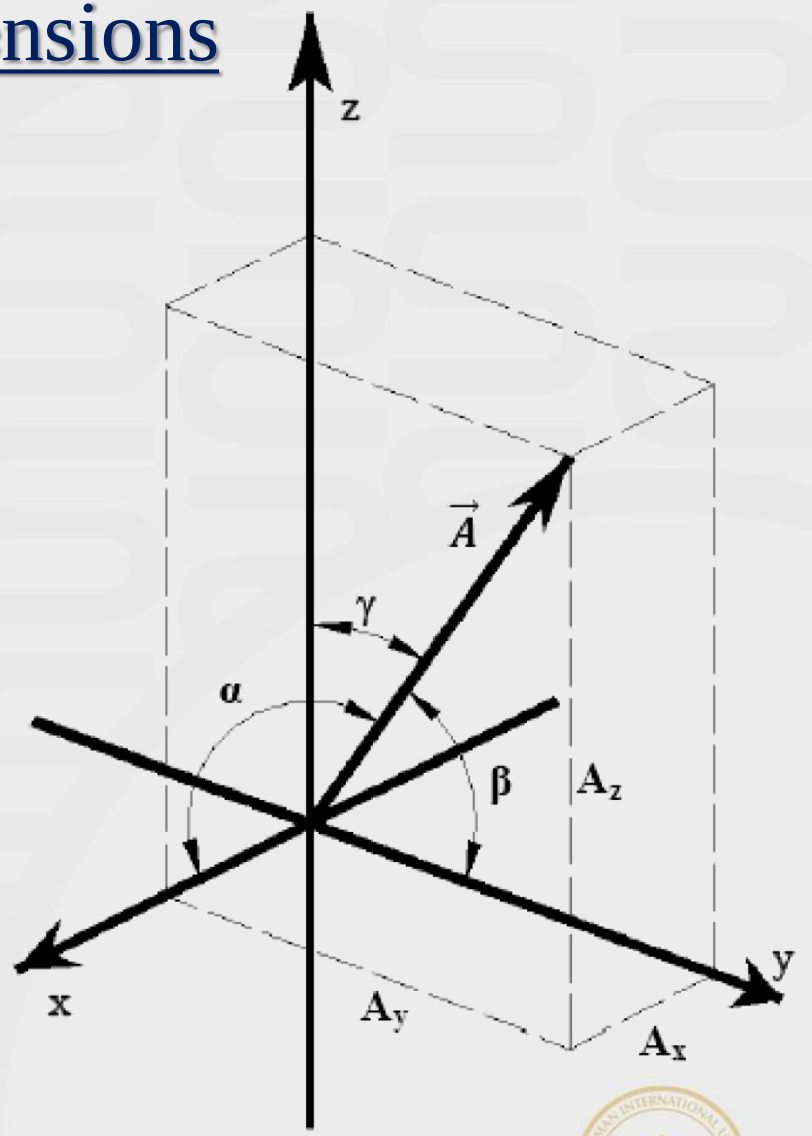
$$\vec{A} = A_x\vec{i} + A_y\vec{j} + A_z\vec{k}$$

The magnitude can be obtained as:

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

The direction of the vector :

α, β and γ.



Vector Components Three-dimensions

Vector Components Three-dimensions α, β and γ

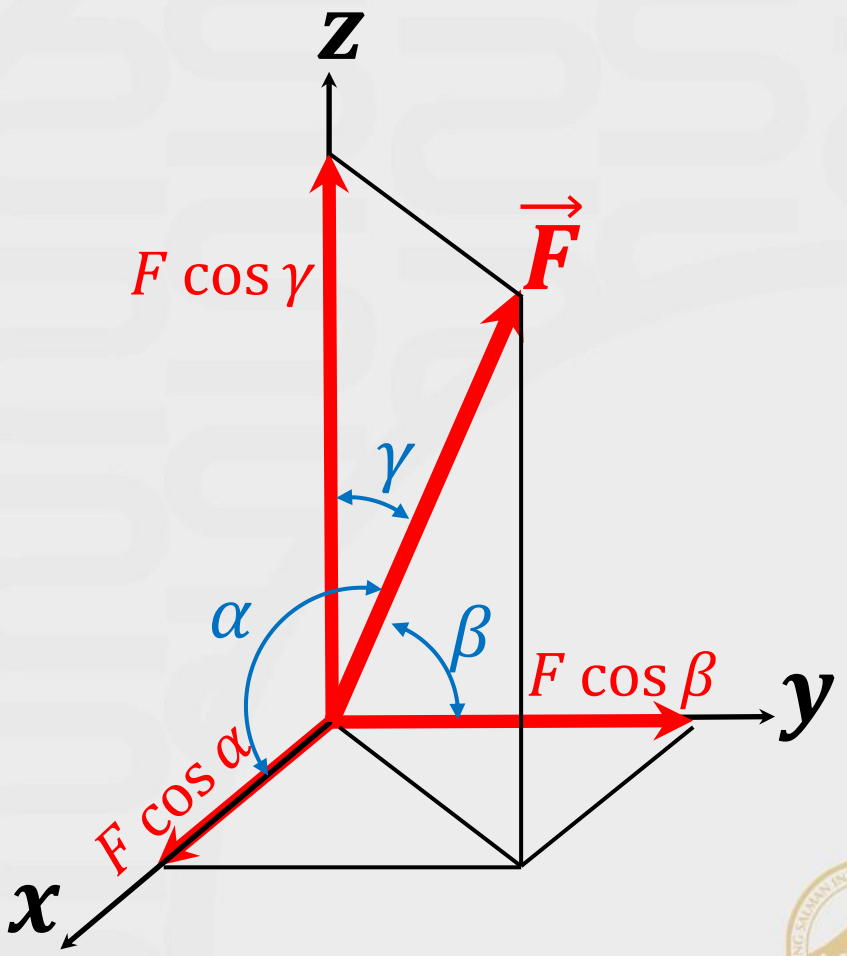
Let $\vec{F}$ be a vector is defined as the coordinate direction angles α, β and γ measured between the tail of $\vec{F}$ and the positive x, y and z axes respectively as shown in Figure.

So:

$$\vec{F} = F\cos\alpha\, \boldsymbol{i} + F\cos\beta\, \boldsymbol{j} + F\cos\gamma\, \boldsymbol{k}$$

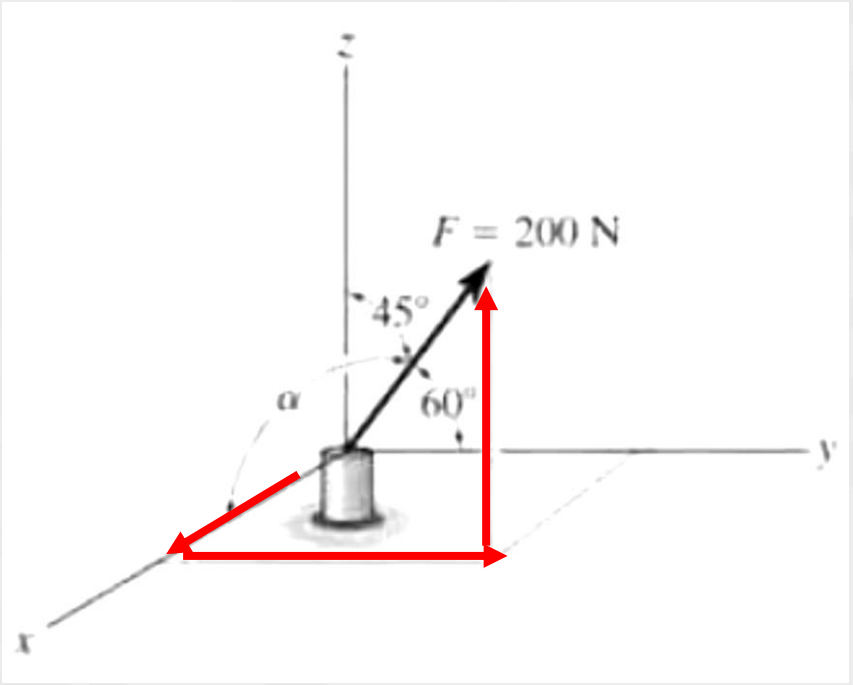
Where

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$



Example 1

Express the force F shown in Figure as a Cartesian vector.



SOLUTION:

$$\cos^2\alpha + cos^2\beta + cos^2\gamma = 1$$

$$\cos^2\alpha + cos^2 60 + cos^2 45 = 1$$

$$\cos\gamma = \sqrt{1 - cos^2 60 - cos^2 45} = \pm 0.5$$

$$\therefore \cos\gamma = +0.5$$

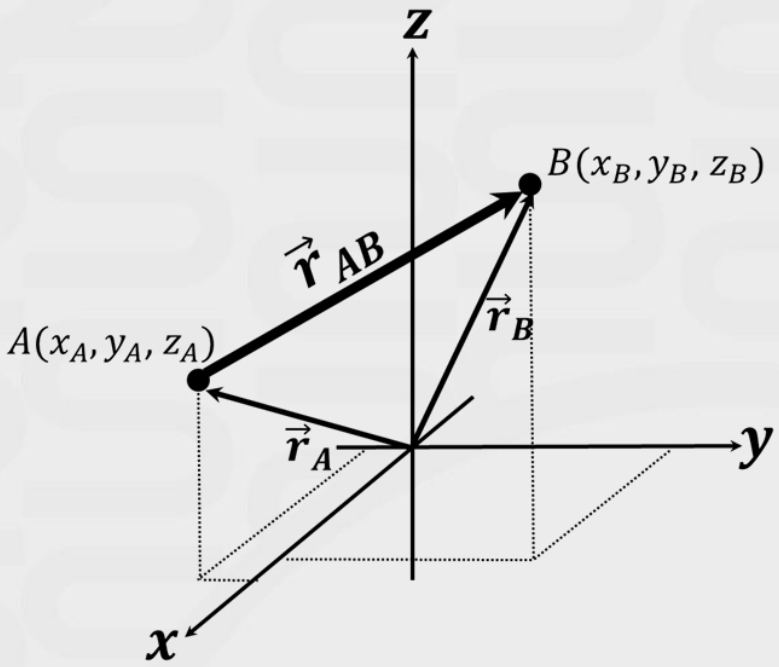
$$\vec{\boldsymbol{F}} = Fcos\,\alpha\,\boldsymbol{i} + Fcos\,\beta\,\boldsymbol{j} + Fcos\,\gamma\,\boldsymbol{k}$$

$$\vec{\boldsymbol{F}} = (200 \times 0.5)\,\boldsymbol{i} + (200\cos 60)\,\boldsymbol{j} + (200\cos 45)\boldsymbol{k}$$

$$\vec{\boldsymbol{F}} = 100\boldsymbol{i} + 100\,\boldsymbol{j} + 141.4\,\boldsymbol{k}$$

Position Vector

More generally, the position vector directed from point $A(x_A, y_A, z_A)$ to point $B(x_B, y_B, z_B)$ as shown in Figure. The position vector $\vec{r}_{AB}$ for that particle is:



$$\vec{r}_{AB} = (x_B - x_A)\,\boldsymbol{i} + (y_B - y_A)\,\boldsymbol{j} + (z_B - z_A)\,\boldsymbol{k}$$

The magnitude of the vector is:

$$r_{AB} = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$$

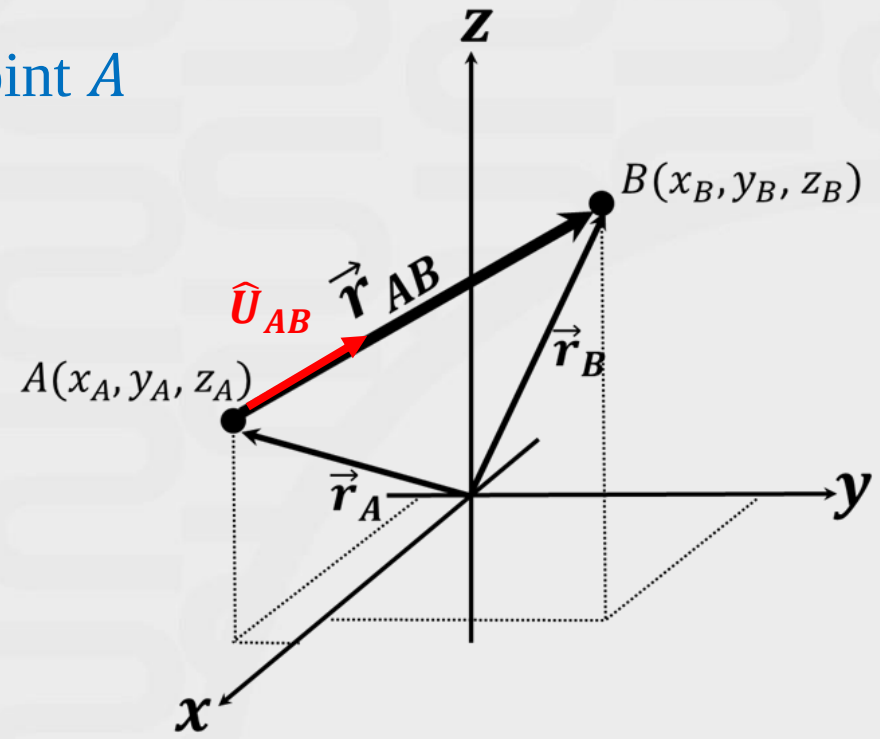
Position Vector

The magnitude of the vector is:

$$r_{AB} = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$$

The unit vector in the direction of the vector $\vec{r}_{AB}$ (from point *A* to point *B*) is:

$$\hat{U}_{AB} = \frac{\vec{r}_{AB}}{r_{AB}}$$

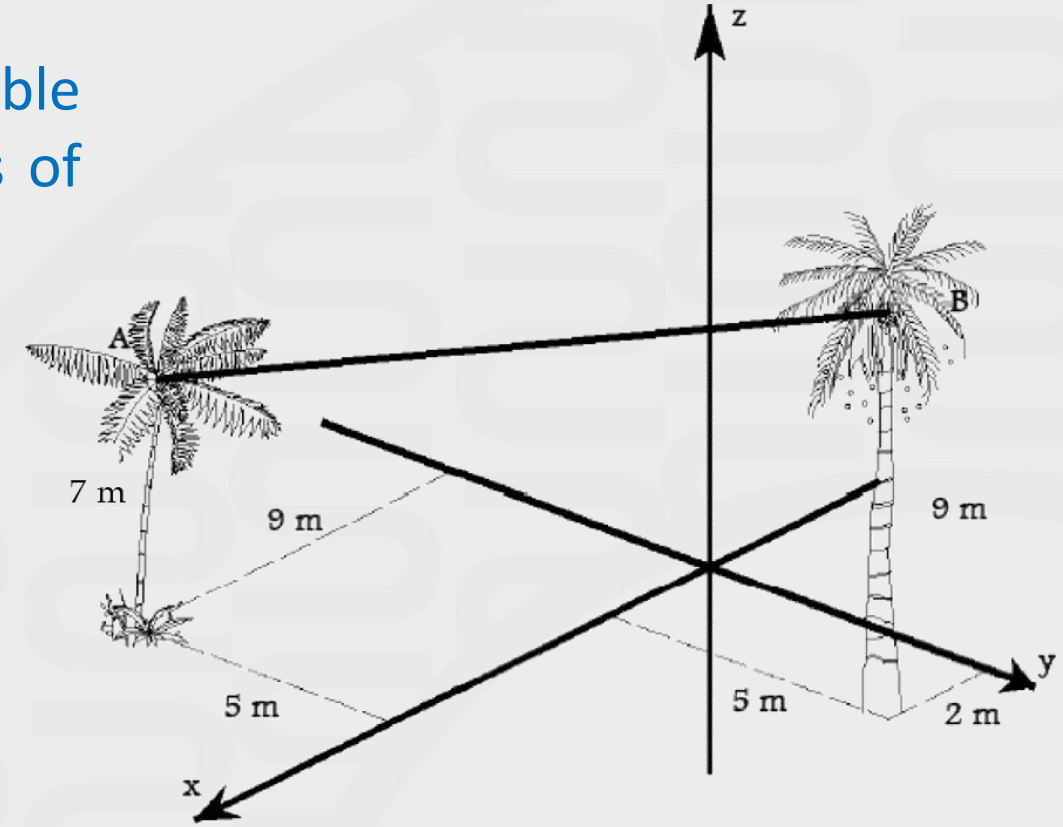


The direction angles (α, β, γ) are obtained as:

$$\cos\alpha = \frac{x_B - x_A}{AB},\ \cos\beta = \frac{y_B - y_A}{AB}\ and\ \cos\gamma = \frac{z_B - z_A}{AB}$$

Example 3

The cable AB connected between two palms. If the cable tension is 700 N , express the tension force in terms of unit vectors $\boldsymbol{i}$,$\boldsymbol{j}$and $\boldsymbol{k}$, acting on point A.



SOLUTION:

$A = (9, -5, 7) \quad B = (2, 5, 9)$

$$\vec{\boldsymbol{r}}_{\boldsymbol{AB}} = (x_B - x_A)\,\boldsymbol{i} + (y_B - y_A)\,\boldsymbol{j} + (z_B - z_A)\,\boldsymbol{k}$$

$$\vec{\boldsymbol{r}}_{\boldsymbol{AB}} = (2 - 9)\,\boldsymbol{i} + \big(5 - (-5)\big)\,\boldsymbol{j} + (9 - 7)\,\boldsymbol{k} = -7\,\boldsymbol{i} + 10\,\boldsymbol{j} + 2\,\boldsymbol{k}$$

$$\vec{r}_{AB} = -7\,\boldsymbol{i} + 10\,\boldsymbol{j} + 2\,\boldsymbol{k}$$

$$r_{AB} = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$$

$$r_{AB} = \sqrt{(-7)^2 + (10)^2 + (2)^2} = 12.4\,m$$

$$\hat{U}_{AB} = \frac{\vec{r}_{AB}}{r_{AB}}$$

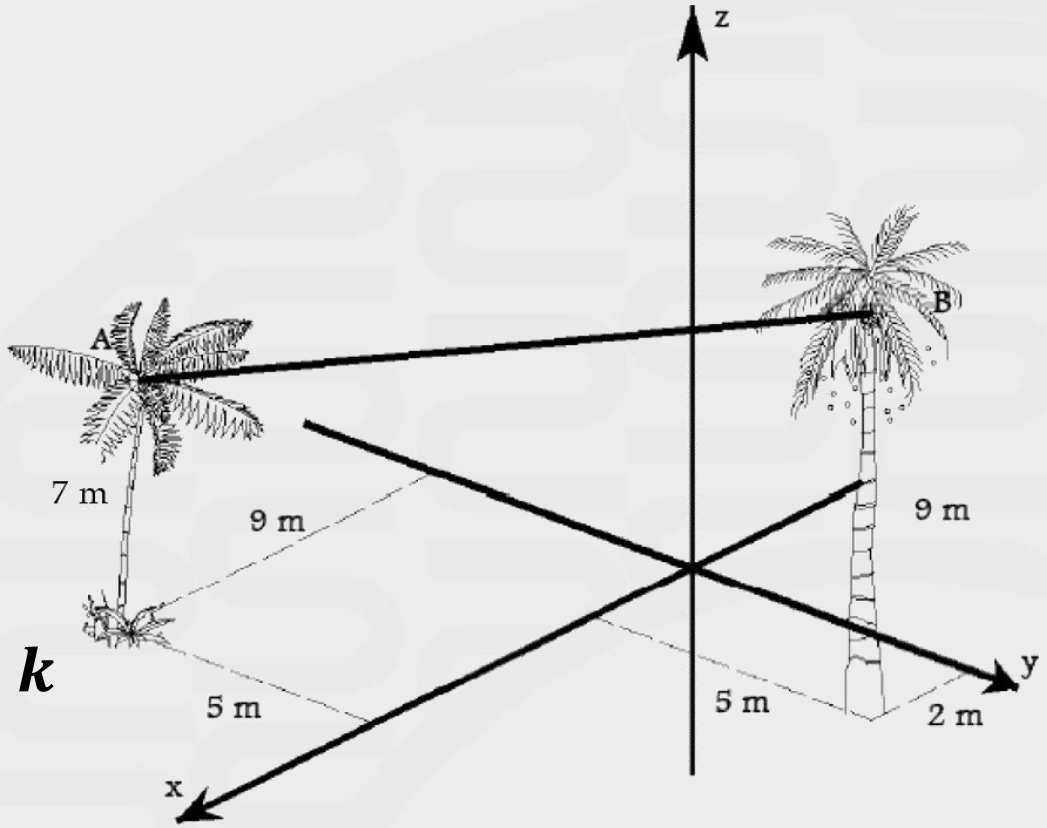
$$\hat{U}_{AB} = \frac{-7\,\boldsymbol{i} + 10\,\boldsymbol{j} + 2\,\boldsymbol{k}}{12.4} = -0.565\,\boldsymbol{i} + 0.808\,\boldsymbol{j} + 0.161\,\boldsymbol{k}$$

$$\vec{T} = T\hat{U}_{AB}$$

$$\vec{T} = 700(-0.565\,\boldsymbol{i} + 0.808\,\boldsymbol{j} + 0.161\,\boldsymbol{k})$$

$$\vec{T} = -395.5\,\boldsymbol{i} + 565.6\,\boldsymbol{j} + 112.7\,\boldsymbol{k}$$

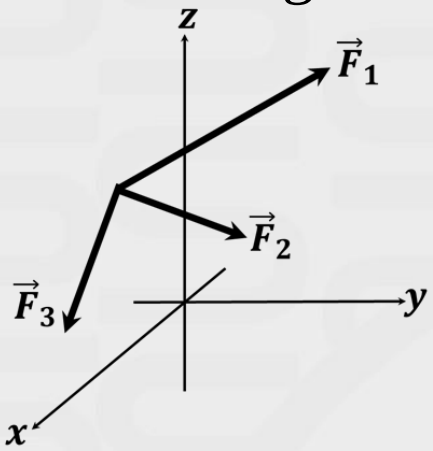
OR $\vec{T} = 700\left(\frac{-7\,\boldsymbol{i} + 10\,\boldsymbol{j} + 2\,\boldsymbol{k}}{\sqrt{(-7)^2 + (10)^2 + (2)^2}}\right)$



Resultant force

Resultant of Coplanar Forces (Three-dimensions)

Consider the three concurrent forces as shown in Figure



$$\vec{F}_1 = F_{1x}\boldsymbol{i} + F_{1y}\boldsymbol{j} + F_{1z}\boldsymbol{k}$$

$$\vec{F}_2 = F_{2x}\boldsymbol{i} + F_{2y}\boldsymbol{j} + F_{2z}\boldsymbol{k}$$

$$\vec{F}_3 = F_{3x}\boldsymbol{i} + F_{3y}\boldsymbol{j} + F_{3z}\boldsymbol{k}$$

$$\vec{R} = \sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

$$\vec{R} = (F_{1x} + F_{2x} + F_{3x})\boldsymbol{i} + (F_{1y} + F_{2y} + F_{3y})\boldsymbol{j} + (F_{1z} + F_{2z} + F_{3z})\boldsymbol{k}$$

$$\vec{R} = R_x\boldsymbol{i} + R_y\boldsymbol{j} + R_z\boldsymbol{k}$$

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2}$$ Magnitude

$$\alpha = \cos^{-1}\frac{R_x}{R}$$

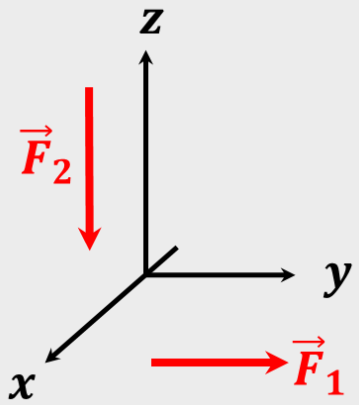
$$\beta = \cos^{-1}\frac{R_y}{R}$$

$$\gamma = \cos^{-1}\frac{R_z}{R}$$

Directiom

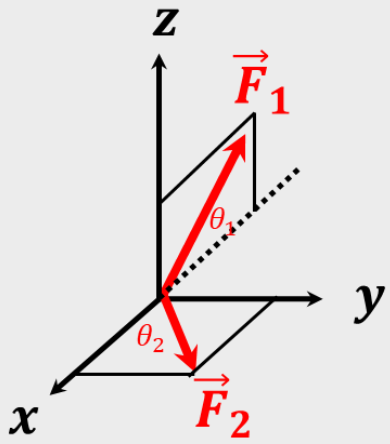
Force Representation

Force Parallel to an axis



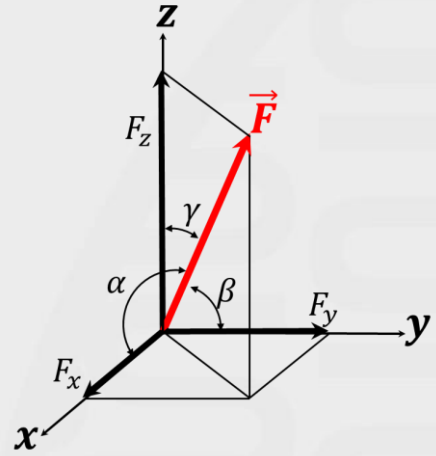
$$\vec{F}_1 = F_1\ \boldsymbol{j}$$
$$\vec{F}_2 = -F_2\ \boldsymbol{k}$$

Force in the plane



$$\vec{F}_1 = -F_1 cos\theta_1\ \boldsymbol{i} + F_1 sin\theta_1 \boldsymbol{k}$$
$$\vec{F}_2 = F_2 cos\theta_2\ \boldsymbol{i} + F_2 sin\theta_2 \boldsymbol{j}$$

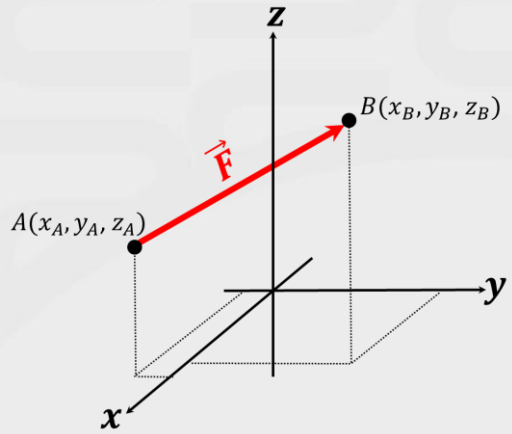
Force inclined by angles α, β and γ



$$\vec{F} = F cos\,\alpha\ \boldsymbol{i} + F cos\,\beta\ \boldsymbol{j} + F cos\,\gamma\ \boldsymbol{k}$$

$$cos^2\alpha + cos^2\beta + cos^2\gamma = 1$$

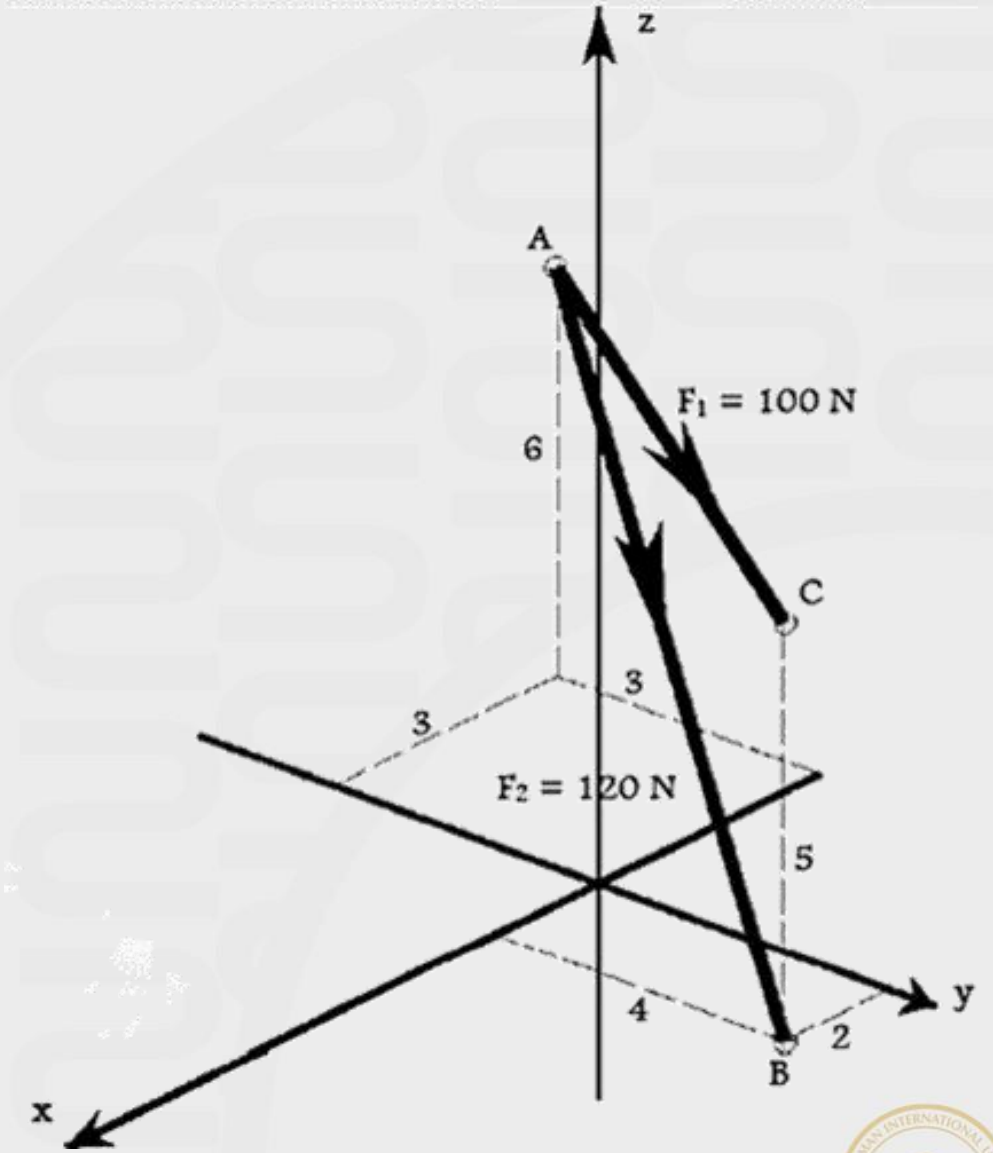
Force Between 2 points



$$\vec{F} = F\left(\frac{(x_B - x_A)\ \boldsymbol{i} + (y_B - y_A)\boldsymbol{j} + (z_B - z_A)\boldsymbol{k}}{\sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}}\right)$$

Example 4

For the shown two forces F1 and F2. Determine the magnitude and direction angles of the resultant of the two forces.



- **For Force F_1:**

$$\because \vec{F}_1 = F_1 \frac{(x_C - x_A)\vec{i} + (y_C - y_A)\vec{j} + (z_C - z_A)\vec{k}}{\sqrt{(x_C - x_A)^2 + (x_C - x_A)^2 + (x_C - x_A)^2}}$$

$$\therefore \vec{F}_1 = 100 \times \left[\frac{(2 - -3)\vec{i} + (4 - -3)\vec{j} + (5 - 6)\vec{k}}{\sqrt{(5)^2 + (7)^2 + (-1)^2}}\right]$$

$$\therefore \vec{F}_1 = 57.735\vec{i} + 80.829\vec{j} - 11.547\vec{k}$$

- **For Force F_2:**

$$\because \vec{F}_2 = F_2 \frac{(x_B - x_A)\vec{i} + (y_B - y_A)\vec{j} + (z_B - z_A)\vec{k}}{\sqrt{(x_B - x_A)^2 + (x_B - x_A)^2 + (x_B - x_A)^2}}$$

$$\therefore \vec{F}_2 = 120 \times \left[\frac{(2 - -3)\vec{i} + (4 - -3)\vec{j} + (0 - 6)\vec{k}}{\sqrt{(5)^2 + (7)^2 + (-6)^2}}\right]$$

$$\therefore \vec{F}_2 = 57.207\vec{i} + 80.09\vec{j} - 68.649\vec{k}$$

$$\because \vec{R} = \sum \vec{F}_i = \vec{F}_1 + \vec{F}_2$$

$$\therefore \vec{R} = (57.735 + 57.207)\vec{\imath} + (80.829 + 80.09)\vec{\jmath} + (-11.547 - 68.649)\vec{k}$$

$$\therefore \vec{R} = 114.942\vec{\imath} + 160.919\vec{\jmath} - 80.196\vec{k}$$

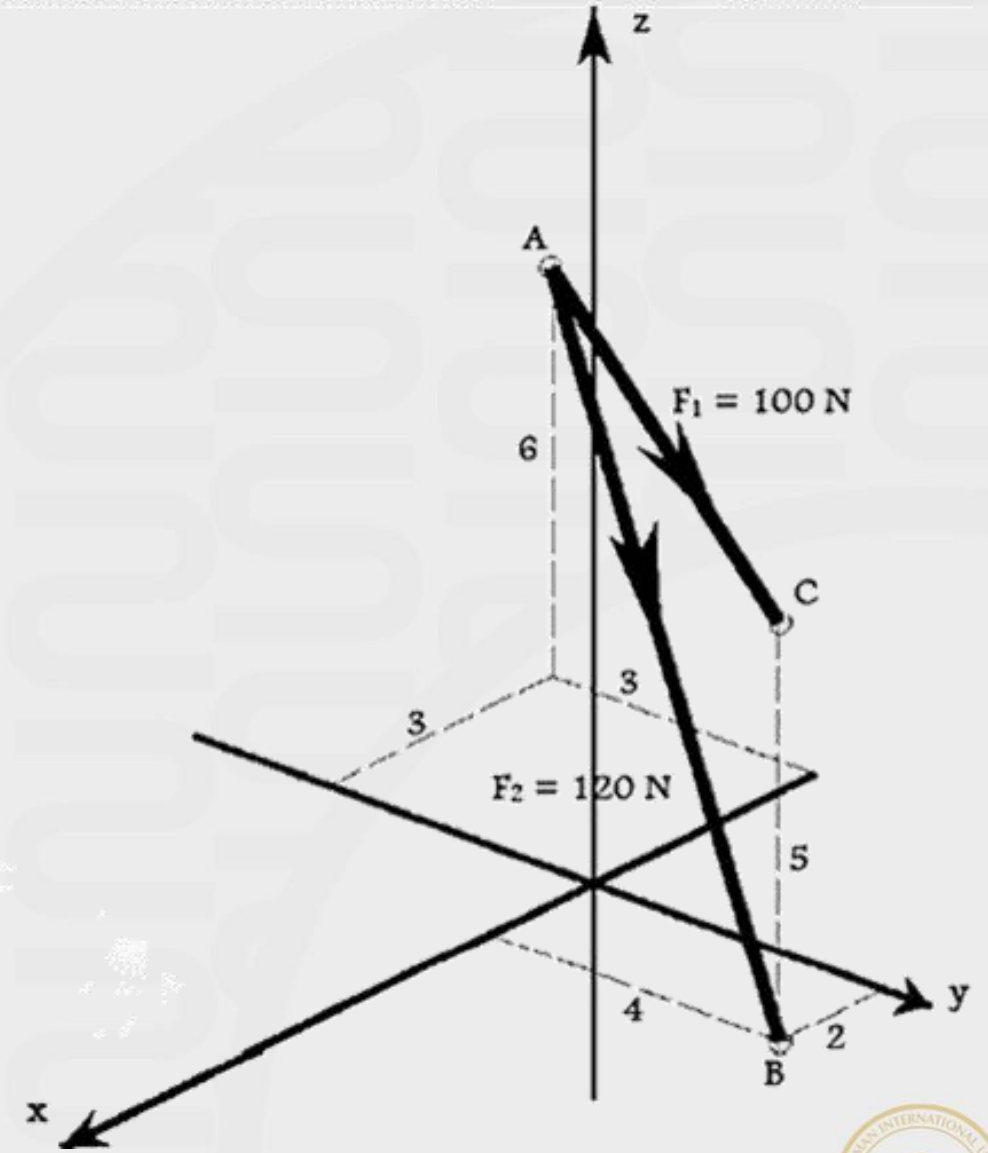
$$\therefore R = \sqrt{114.942^2 + 160.919^2 + 80.196^2}$$

$$\therefore R = 213.396\ N$$

$$\therefore \cos\alpha = \frac{114.942}{213.396} \quad \Longrightarrow \quad \alpha = 57.409^o$$

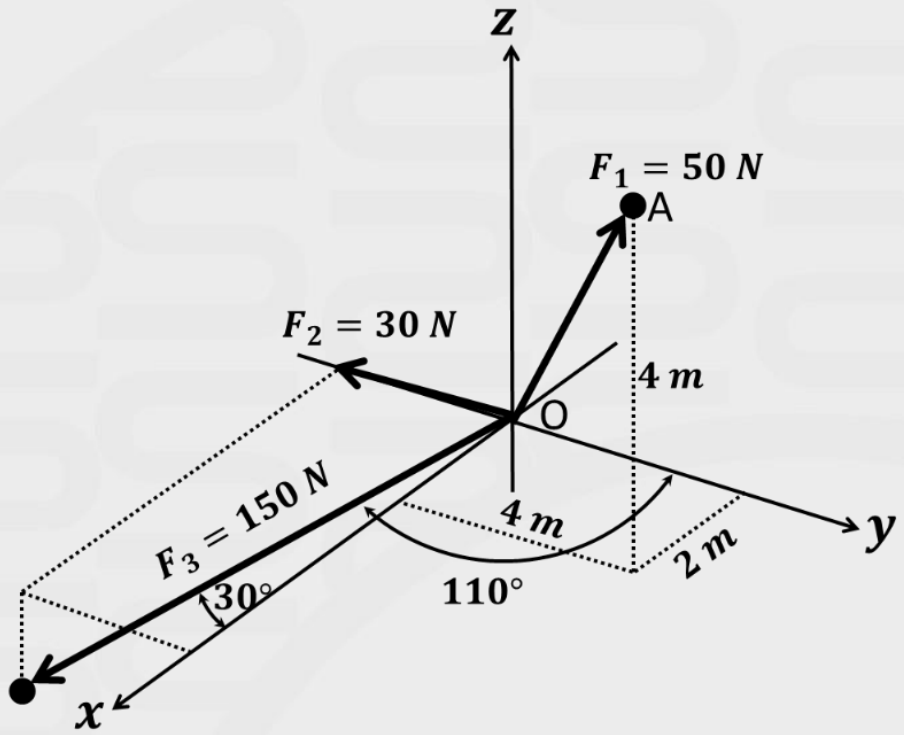
$$\therefore \cos\beta = \frac{160.919}{213.396} \quad \Longrightarrow \quad \beta = 41.054^o$$

$$\therefore \cos\gamma = \frac{-80.196}{213.396} \quad \Longrightarrow \quad \gamma = 112.074^o$$



Example 4

Three forces act on the hook at point O as shown in Figure. Determine the magnitude and direction of the resultant force.



SOLUTION:

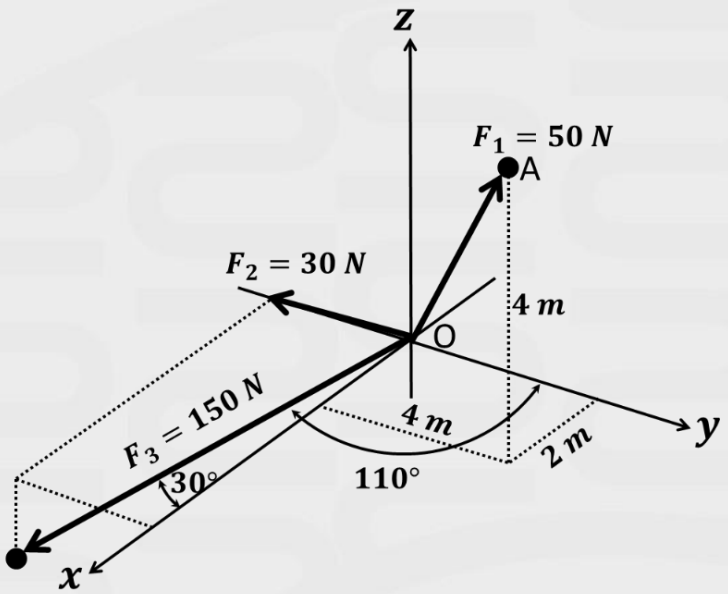
$$\vec{F}_1 = F_1 \hat{U} = F_1 \frac{\vec{r}}{r} = F_1 \left[\frac{x\,i + y\,j + z\,k}{\sqrt{x^2+y^2+z^2}}\right]$$

$$\vec{F}_1 = 50 \left[\frac{2\,i+4\,j+4\,k}{\sqrt{2^2+4^2+4^2}}\right]$$

$$\vec{F}_1 = 16.67\ i + 33.3\ j + 33.3\ k$$

$$\vec{F}_2 = -30\ j$$

Example 4



$$\cos^2 30 + cos^2 110 + cos^2 \gamma = 1$$

$$\cos\gamma = \sqrt{1 - cos^2 30 - cos^2 110} = -0.364?$$

$$\vec{F_3} = 150cos\ 30\ \boldsymbol{i} + 150cos\ 110\ \boldsymbol{j} + 150cos\ \gamma\ \boldsymbol{k}$$

$$\vec{F_3} = 129.9\ \boldsymbol{i} - 51.3\ \boldsymbol{j} - 54.6\ \boldsymbol{k}$$

$$\vec{R} = \sum \vec{F} = \vec{F_1} + \vec{F_2} + \vec{F_3}$$

$$\vec{R} = 146.6\ \boldsymbol{i} - 48\ \boldsymbol{j} - 21.2\ \boldsymbol{k}$$

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2} = \sqrt{(146.6)^2 + (48)^2 + (21.2)^2} = \underline{155.7\ N}$$

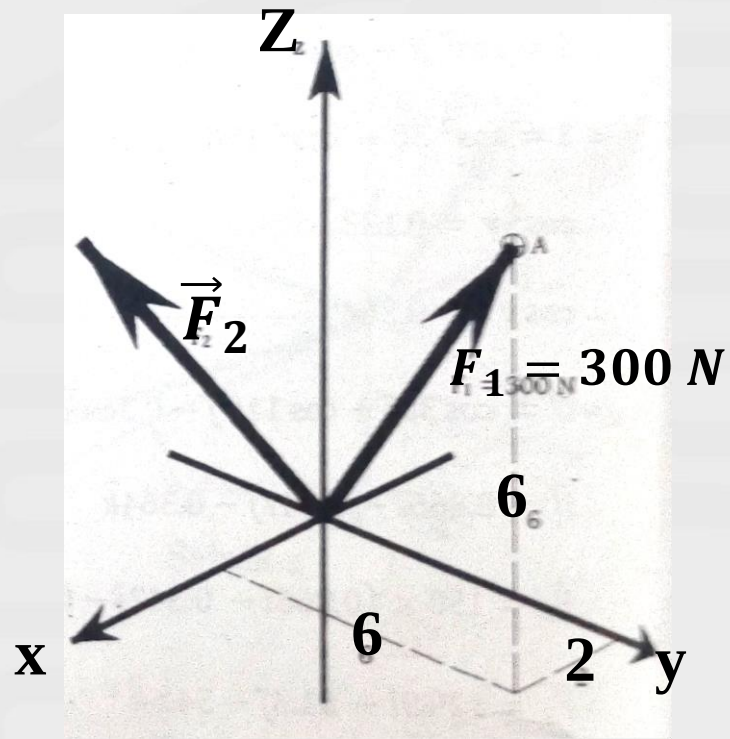
$$\alpha = \cos^{-1}\frac{R_x}{R} = \cos^{-1}\frac{146.6}{155.7} = \underline{19.7°}$$

$$\beta = \cos^{-1}\frac{R_y}{R} = \cos^{-1}\frac{-48}{155.7} = \underline{107.9°}$$

$$\gamma = \cos^{-1}\frac{R_z}{R} = \cos^{-1}\frac{-21.2}{155.7} = \underline{97.9°}$$

Example 5

If the resultant of the two forces is $\vec{R} = 500\,\vec{k}$ N. Determine the magnitude and direction of F_2 .



SOLUTION:

$$\vec{F}_1 = F_1\hat{U} = F_1\frac{\vec{r}}{r} = F_1\left[\frac{x\,i + y\,j + z\,k}{\sqrt{x^2+y^2+z^2}}\right]$$

$$\vec{F}_1 = 300\left[\frac{2\,i+6\,j+6\,k}{\sqrt{2^2+6^2+6^2}}\right]$$

$$\vec{F}_1 = 68.8\ i + 206.5\ j + 206.5\ k$$

$$\vec{F}_2 = F_{2x}\,i + F_{2y}\,j + F_{2z}\,k$$

$$\vec{R} = \sum \vec{F} = \vec{F}_1 + \vec{F}_2$$

Example 5

$\vec{i}$ $\Rightarrow$ $0 = 68.8 + F_{2x}$ $\Rightarrow$ $F_{2x} = -68.8\ N$

$\vec{j}$ $\Rightarrow$ $0 = 206.5 + F_{2y}$ $\Rightarrow$ $F_{2y} = -206.5\ N$

$\vec{k}$ $\Rightarrow$ $500 = 206.5 + F_{2z}$ $\Rightarrow$ $F_{2z} = 293.5\ N$

$$F_2 = \sqrt{F_{2x}^2 + F_{2y}^2 + F_{2z}^2} = \sqrt{(68.8)^2 + (206.5)^2 + (293.5)^2}$$

$$F_2 = \underline{365.4\ N}$$

$$\alpha = \cos^{-1}\frac{F_{2x}}{F_2} = \cos^{-1}\frac{-68.8}{365.4} = \underline{100.9°}$$

$$\beta = \cos^{-1}\frac{F_{2y}}{F_2} = \cos^{-1}\frac{-206.5}{365.4} = \underline{124.4°}$$

$$\gamma = \cos^{-1}\frac{F_{2z}}{F_2} = \cos^{-1}\frac{293.5}{365.4} = \underline{35.6°}$$

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